

gravity and magnetic fields, and more!

Nokia's Mobile Augmented Reality Applications (MARA) project uses camera-equipped mobile devices for "video see-through mobile AR," meaning that it can overlay information right on the cell phone's screen, depending on what the camera is being pointed at. We learnt from <http://tinyurl.com/32tmkx> about such a prototype phone: it is equipped with MARA software, and hardware including a GPS, an accelerometer, and a compass. Conceptually, it's simple. As the phone captures a video stream, MARA collates information from the aforementioned hardware to pinpoint the location and orientation of the phone. It then looks in a database to determine what objects *should* be visible when the camera is at that location. (The database could be on the

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with any character using the 'augmented people' module."

Then, from <http://tinyurl.com/35jpxo>, we learn that a system has been developed for tourists at sites such as Pompeii to be able to see the people who lived there before the eruption of Vesuvius destroyed the city. They're calling it a "break-through prototype AR system"; it can add "digital people" and other computer-generated, animated elements in real-time to what you "actually" see. Again, "actually" is in quotes, because... oh, well.

Turns out someone's actually designed an AR—read VISTA—version of Quake, in which the monsters are overlaid on the real scene in front of you. Read about the ARQuake Project—and watch the videos—at <http://tinyurl.com/c6459>.

We said it, and we'll say it again—the applications are limitless. The US military has been doing VISTA-oriented research for many years now. The possibilities? How about this one: there is data on the construction of a building, with its entrances and exits; a soldier with a Head-Mounted Display would see all these at once instead of just the one entrance he's facing. Then, think about training: soldiers could "face" troops and tanks and all, while actually situated on real ground, and perform the appropriate actions.



Just one aspect of the AR problem. User points AR device at object; device collates lots of measurements; edges are detected. Even this is hard!

phone, or accessed remotely.) Some pattern matching is involved here, and once it's been established what objects are being viewed by the camera, MARA highlights them and provides information about them (and hyperlinks, if available in the database). You can imagine the rest: if a restaurant is being viewed and is also in the database, MARA could (potentially) display the menu and wait time. If there were a hyperlink as well, you'd be able to visit the restaurant's Web site!

Watch the videos at <http://tinyurl.com/y883yj>.

VISTA: The Middle Ground

What we were calling VISTA (Virtual Imagery Superimposed on Top of Actuality)—the infusion of virtual elements into a real scene—is, as we said, something like mixed reality. Strictly speaking, according to many people's definitions, this is AR—because it's all about augmenting a scene. We've coined the term VISTA because we think AR *enhances* the reality you're experiencing, while VISTA is *adding virtual elements* that shift focus away from the "reality."

Consider these useful applications:

- An interior designer can have a view of the before-and-after scene. He sees a room as it is, and when he presses a button on his Head-Mounted Display, his planned additions—like a sofa—appear as though they were in the room.
- Here's a gem of an idea we found on the Net: "Don't like your office environment? Add plants, waterfalls, and hummingbirds to your office with the 'sounds of nature' software module. Is your significant other unbearably ugly? Overlay their natural face

Closing Thoughts

Some aver that AR is a potentially dangerous technology. When it becomes commonplace, you'd have to get your "broadcasts" from somewhere—for example, we said ad agencies could send their information to the information servers so that the ads get displayed along with tourist-aiding tags. Now if those broadcasts are to be received from some kind of central agency—either a government, or a large corporation—it *could* result in the ultimate in mind control. The reality you perceive would be controlled by someone else! If the idea is not clear, think about how much of our minds is controlled by Google today. With that as a real-life example, think about "subscribing" to AR channels, and how they'd influence your perception of reality.

Continuing along the lines of "subscribing to channels," think of how exactly you could *use* AR. As a tourist in Kenya, you'd subscribe to the Kenyan tourism broadcast. On a trek, you'd tune in to "Nature"—and trees and shrubs and little furry animals could get tagged with their names. Just for fun, at a high-profile business meeting, you'd be able to figure how expensive the business suits around you are. Talking to someone who doesn't speak English? Use translation AR, and you'd "listen" to the person in English.

We'll leave the fantasising to you. But we need to reinforce that AR and VISTA aren't fads; research is red-hot and people are dead-serious about it. (Please remember, only we coined "VISTA") But it might just lead to nothing much, going the way of VR—if you remember how wild people were about VR a decade and a half ago.

Ultimately, it's about *potential*. To use a cliché and an obvious phrase, we'll end by saying that Augmented Reality has the potential to fundamentally change our life-experience. ■

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